Lecture 39 — Least squares, data fitting, and positive definiteness

Wed Dec 2

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Boyd video (optional): Lecture 34 — least squares examples · Lecture 35 — least-squares data fitting · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, and Lachniet Ch. 2 (least-squares / data-fitting exercises). Related objectives: Lectures. Schedule.

Bring paper and a laptop if you have one. Warm-up, fill-in, a small real dataset on the board/software, then **Your turn**. Full least-squares software practice is Lab 6 on **Friday** (same day as Quiz 10 — Quiz 10 is Week 13 eigenvalues, **not** today's fitting).

Learning objectives

By the end of class, students should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A.
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that $A^\mathsf{T}A$ is symmetric positive semidefinite, explain $x^\mathsf{T}A^\mathsf{T}Ax = \|Ax\|^2$, and recognize that $A^\mathsf{T}A$ is positive definite when A has linearly independent columns.

Warm-up

Monday's setup: minimize $\|Ax - b\|$. At a minimizer $\hat{x}$, the residual $r = b - A\hat{x}$ is orthogonal to $C(A)$, which is equivalent to the **normal equations**

$$A^\mathsf{T}A\,\hat{x} = A^\mathsf{T}b.$$

(a) If A is $m \times n$ with $m > n$, what size is $A^\mathsf{T}A$? Why is that nicer than solving $Ax = b$ directly?

(b) True or false: $A^\mathsf{T}A$ is always symmetric.

Fill in

Normal equations / software. Solving $A^\mathsf{T}A\hat{x} = A^\mathsf{T}b$ (when A has independent columns) gives the unique least-squares $\hat{x}$. In practice we often call a solver instead of forming $A^\mathsf{T}A$ by hand:

Task	Octave	Python
Least-squares $\hat{x}$	`xhat = A \ b`	`xhat = np.linalg.lstsq(A, b, rcond=None)[0]`
Fitted values	`bhat = A*xhat`	`bhat = A @ xhat`
Residual	`r = b - bhat`	`r = b - bhat`
RMS	`rms = norm(r)/sqrt(m)`	`rms = np.linalg.norm(r)/np.sqrt(m)`

Orthogonality check. At a least-squares solution, $A^\mathsf{T} r \approx$ _____ (numerical noise ok).

RMS. Root-mean-square residual RMS $= \|r\|/\sqrt{m}$ reports a typical per-equation error in the same units as b.

Useful vs coefficients. Always ask: do fitted values track the data? Is RMS small *in context*? Blindly quoting $\hat{x}$ is not enough.

Positive semidefinite / definite (Boyd / class notes). A symmetric matrix P is **positive semidefinite (PSD)** if $x^\mathsf{T} P x \geq 0$ for all x; **positive definite (PD)** if $x^\mathsf{T} P x > 0$ for all $x \neq 0$.

For any A:

$$x^\mathsf{T}(A^\mathsf{T} A)x = \|Ax\|^2 \geq 0,$$

so $A^\mathsf{T} A$ is always _____. It is **positive definite** precisely when $Ax = 0 \Rightarrow x = 0$, i.e. when A has _____ columns — then the normal equations have a unique solution.

Worked in class

W1. Tiny fit by hand / software. Four campus shuttle wait times (minutes) at hours $t = 1, 2, 3, 4$ after 8am:

$$b = (6,\ 7,\ 10,\ 11).$$

Model a line $b \approx x_1 + x_2 t$ with

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

(a) Form $A^\mathsf{T} A$ and $A^\mathsf{T} b$ (or use `A\b` / `lstsq`).
(b) Report $\hat{x} = (x_1, x_2)$ and the fitted values $\hat{b} = A\hat{x}$.
(c) Compute r and RMS. In one sentence: is a linear trend reasonable here?

Expected (for checking): $\hat{x} = (4,\ 1.8)$ exactly from the normal equations; $\hat{b} = (5.8,\ 7.6,\ 9.4,\ 11.2)$; RMS $= \sqrt{0.2} \approx 0.45$ minutes.

W2. Orthogonality. With your $\hat{x}$ from W1, compute $A^{\mathsf{T}} r$. Should it be near $(0, 0)$?

W3. Plot story. On axes "hour t" vs "wait (min)", sketch data points and the fitted line. Where is the residual for $t = 1$ visible?

W4. PSD identity. Expand $x^{\mathsf{T}} A^{\mathsf{T}} A x$ for $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and a general $x = (x_1, x_2)$. Confirm it equals $\|Ax\|^2$ and is ≥ 0.

W5. PD vs PSD. Explain why $A^{\mathsf{T}} A$ fails to be positive definite if the two columns of A are equal. What goes wrong for uniqueness of $\hat{x}$?

Your turn

Y1. Using the W1 data, suppose software returns $\hat{x} = (4,\ 1.8)$. Compute $\hat{b}_1 = x_1 + x_2 \cdot 1$ and the first residual $r_1 = b_1 - \hat{b}_1$.

Y2. True or false: $A^{\mathsf{T}}A$ is always positive definite for every A.

Y3. Why check $A^{\mathsf{T}}r \approx 0$ after a least-squares solve?

Y4. A fit has RMS = 0.02 °C for outdoor temperature data that typically varies by 10 °C across the day. Useful model, or not? One sentence.

Answers (Your turn)

Y1. $\hat{b}_1 = 4 + 1.8 = 5.8$; $r_1 = 6 - 5.8 = 0.2$.

Y2. **False.** $A^\mathsf{T}A$ is always PSD, but PD only when A has linearly independent columns (equivalently $\ker A = \{0\}$).

Y3. It is the normal-equation / geometric check that the residual is orthogonal to $C(A)$ — evidence that $\hat{x}$ really is a least-squares critical point (up to roundoff).

Y4. **Useful** in context: RMS $\ll$ the natural scale of the data, so typical pointwise error is tiny compared with day-scale variation.